

Final Project Guideline

The final project builds upon the approved proposal and aims to demonstrate your ability to execute a complete data science project using Python and Hugging Face resources. The merit of your final project will be demonstrated and evaluated through a GitHub repository which contains:

1. A well-documented executable codebase.
2. A 2-page summary report.

*An average final project should take about 40 hours.

Project Scope

- Use data analysis/science techniques taught in this class.
- Focus on practical application of Python tools

Report Guideline

- Deadline: 12/04/24
- The final project report should be in the format of a 2-page IEEE conference paper. [\[template\]](#)
- Sections:

Introduction

- Provide background and motivation of the project.
- State a concrete project goal.
- Conduct a comprehensive review of existing literature, focusing on recent advancements and state-of-the-art.

Method

- Problem formulation: learning problem formulation (input, output), dataset description and model formulation.
- Walk through the methodologies used to solve the problem.

Results

- Data pipeline or model set up.
- Figures that present numerical simulation results.
- Interpretation of the results.

Conclusion

- Tips for the final report:
 - Writing is a great tool to organize your thought. Write a draft report early and use it to lay out your project plan.
 - Start from the Method and Results sections, then the Introduction and Related Work sections. Write the Abstract last.
 - Set aside the first draft for a while and then come back with fresh perspective!
 - Revise the manuscript.

Final Submission

- Deadline: 12/04/24
- Submission Content
 - A summary report (PDF)
 - Code (link to GitHub repository in TXT)

Sample Projects:

1. Preprocess the data and fine-tune a pretrained model on any GLUE task.

[\[Code\]](#)

Potential Extensions:

- Experiment with different model architectures
- Add custom preprocessing steps
- Implement domain-specific optimizations
- Create novel evaluation metrics

2. Fine-tune an image model

[\[Code\]](#)

Potential Extensions:

- Implement custom data augmentation
- Add multi-task capabilities
- Create specialized visualization tools
- Build real-world applications

Some other working notebooks can be found [here](#).

You might need to ensure you have the required packages installed for each notebook (use Claude or ChatGPT or go to office hours for help). You can choose to:

- Use these notebooks as starting points and build upon them
- Focus on in-depth data analysis
- Explore different model architectures
- Create entirely new implementations
- Combine multiple approaches in novel ways
- Follow UM GenAI guideline in: <https://genai.umich.edu/resources/students>

Remember, these examples are meant to inspire rather than limit. Feel free to explore other datasets, models, or approaches that align with your interests while demonstrating your Python data science skills.